

VINELAND STATION RCS, ON, Canada

WMO: 711710

Lat: 43.191N

Lon: 79.399W

Elev: 260

StdP: 14.56

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.0	8.3	-5.4	4.2	5.7	-1.4	5.1	9.6	26.5	24.5	24.0	22.8	10.6	240	0.511

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.8	88.2	72.4	85.3	71.1	82.3	69.7	75.1	84.1	73.5	81.6	72.0	79.2	11.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.3	120.7	80.0	70.8	114.4	78.4	69.3	108.8	77.0	38.8	84.3	37.2	81.8	35.9	79.5	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.3	19.8	17.7	DB	-0.9	92.3	5.1	2.9	-4.6	94.4	-7.6	96.1	-10.5	97.8	-14.2	99.9
			WB	-1.8	78.4	4.9	2.3	-5.3	80.1	-8.2	81.5	-10.9	82.8	-14.5	84.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.0	25.9	26.8	34.4	44.5	55.5	66.0	72.2	70.8	64.4	52.4	41.6	32.3
	DBStd	18.12	10.67	9.57	9.10	8.04	8.25	7.14	5.47	5.16	6.64	7.89	7.85	8.11
	HDD50	3006	748	651	489	204	31	0	0	0	0	63	271	549
	HDD65	6457	1211	1071	950	615	319	73	5	9	90	398	703	1013
	CDD50	2653	2	0	5	40	201	481	689	644	433	139	18	1
	CDD65	626	0	0	0	1	23	104	229	187	73	9	0	0
	CDH74	4224	0	0	0	21	214	764	1632	1119	431	43	0	0
	CDH80	1151	0	0	0	2	54	237	484	266	104	4	0	0
Wind	WSAvg	8.0	10.2	9.9	9.1	8.5	6.8	6.6	6.4	6.3	6.7	8.0	8.7	9.4
Precipitation	PrecAvg	33.8	2.8	2.2	2.5	3.2	2.9	3.0	2.9	2.8	3.1	3.1	2.8	2.8
	PrecMax	41.0	4.8	4.7	4.2	5.2	6.1	7.7	7.5	5.1	7.7	6.5	5.2	5.9
	PrecMin	25.7	1.5	1.1	0.7	1.4	0.8	0.8	0.6	1.0	1.1	1.3	0.9	1.2
	PrecStd	4.5	0.9	0.9	1.1	1.1	1.4	1.6	1.5	1.4	1.4	1.3	1.1	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.2	56.6	63.3	77.1	85.7	90.1	92.4	90.0	87.9	78.9	66.1	57.1
		MCWB	53.4	49.5	54.9	61.5	70.6	71.9	74.0	74.1	72.1	67.8	57.6	52.4
	2%	DB	50.7	49.1	57.5	69.5	80.2	86.5	88.5	85.9	83.0	73.3	61.6	51.5
		MCWB	46.7	44.8	50.5	57.3	66.6	70.5	73.0	72.0	70.5	64.0	54.5	47.1
	5%	DB	44.9	42.4	52.3	63.5	74.6	82.2	85.4	82.8	78.7	68.5	57.9	46.6
		MCWB	40.7	38.0	46.0	53.8	62.8	69.2	71.2	70.6	68.8	60.6	51.3	42.6
	10%	DB	39.7	38.5	46.3	57.3	69.7	78.1	82.1	80.1	75.4	64.6	53.9	42.9
		MCWB	36.8	35.2	42.2	49.4	60.4	67.4	69.8	69.5	67.4	58.6	48.8	39.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.8	51.8	56.1	64.3	71.3	75.0	78.3	76.8	74.9	69.9	59.3	53.3
		MCDB	57.9	55.4	62.4	75.0	83.0	85.9	87.6	85.0	83.5	77.0	63.6	56.1
	2%	WB	47.2	44.5	51.4	59.2	68.1	72.6	75.6	74.5	72.6	65.7	55.3	47.8
		MCDB	50.2	49.6	55.3	67.3	79.0	82.8	84.5	82.8	80.0	72.2	60.3	51.0
	5%	WB	41.4	38.7	46.5	55.0	64.9	70.7	73.7	72.8	70.4	61.9	51.8	43.4
		MCDB	44.2	41.6	51.7	63.1	72.5	79.4	81.6	79.9	76.7	66.5	56.7	46.3
	10%	WB	36.9	35.3	41.8	49.6	61.5	68.6	72.0	71.4	68.3	59.4	49.0	39.9
		MCDB	39.5	38.2	45.8	56.0	68.0	76.3	79.1	77.6	74.3	63.9	53.5	42.6
Mean Daily Temperature Range	5% DB	MDBR	11.7	12.1	12.8	15.0	17.0	15.9	15.8	15.1	15.3	13.3	12.5	10.0
		MCDBR	16.3	18.4	21.2	26.9	24.8	21.5	19.2	18.4	19.3	19.0	19.4	15.0
		MCWBR	14.8	15.7	15.5	17.1	14.2	10.8	9.2	8.7	10.2	12.3	14.5	13.3
	5% WB	MCDBR	15.5	17.7	20.2	25.2	24.0	19.6	16.2	15.5	16.8	16.0	17.7	14.8
		MCWBR	14.5	15.7	15.6	18.0	15.2	11.3	9.0	8.4	9.9	11.7	14.3	13.6
Clear-Sky Solar Irradiance	taub	0.300	0.332	0.341	0.385	0.412	0.444	0.441	0.425	0.390	0.368	0.318	0.303	
	taud	2.424	2.320	2.389	2.301	2.276	2.220	2.253	2.284	2.360	2.394	2.501	2.462	
	Ebn at Noon	265	272	285	279	273	264	264	264	265	256	257	253	
	Edh at Noon	24	31	33	39	42	44	42	40	34	29	22	21	
All-Sky Solar Radiation	RadAvg	435	674	1059	1401	1712	1860	1868	1650	1316	797	498	347	
	RadStd	39	60	115	149	180	136	140	87	120	73	66	40	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page